Analyzing U.S. Yearly Crop Yield by Climate Indicators

From 2010 – 2025

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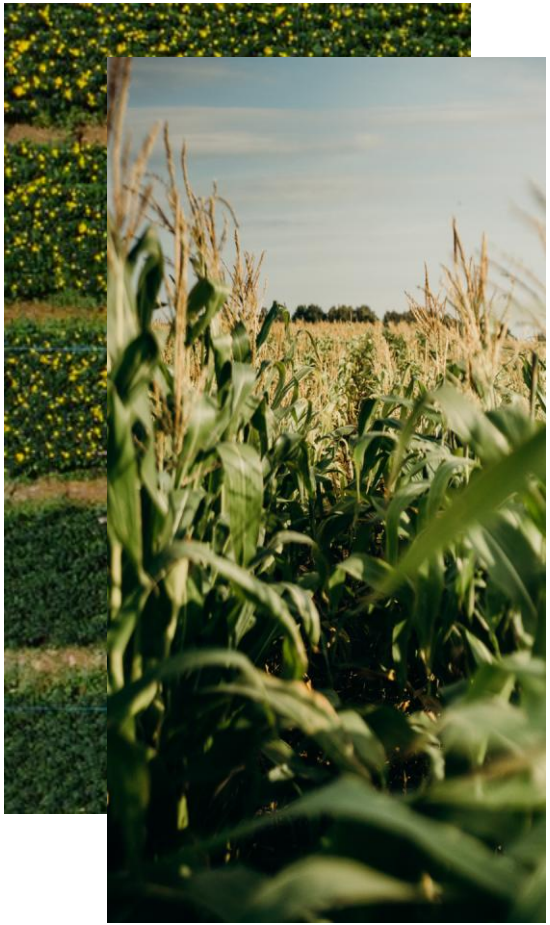
Introduction

Farmers have cited that global warming has led to longer drought, heat waves, excessive rainfall, and erratic weather patterns that have reduced crop yield. In fact, more than 50% of historic crop variability can be attributed due to weather fluctuations (Frieler et al., 2017).

This project runs a **multivariate regression** for yearly (2010-2025) crop yields (bushels/acre) for corn, soybeans, and wheat which are highly critical crops in the United States. The model will isolate the effect that core climate predictors, including annual average temperature, temperature range, and rainfall have on yield for all three crops, controlling for states and year.

Additionally, a **sentiment analysis** will depict farmer's sentiment on each crop's condition in this timeframe.



Data Sources

Data Source	Description	Retrieval/Merging	Data Scale (Post-Cleaning)	Analysis Overview
USDA National Agricultural Statistics Service (NASS)	Contains annual crops and yield (bushels/acre) by state and year	Downloaded as CSV → Stored as SQL table in agclimate.db → merged by state/year with NOAA data as JSON → CSVs for analysis	**601 rows** of data points in merged_agclimate.json file after merging on state/year: *224 rows for corn, 144 rows for soybeans, and 233 rows for wheat*	Contains outcome (yield) and commodities (corn/wheat/soy) used in regression
NOAA U.S. Climate Grid Dataset	Contains annual average, minimum, and maximum temperature and rainfall by state and year	Downloaded 4 .TXT files from NOAA → Converted weekly to annual data → Stored as SQL table in agclimate.db → merged by state/year with USDA data as JSON→ CSVs for analysis	**601 rows** of data points in merged_agclimate.json file after merging on state/year	Contains climate predictors – annual average temperature, max/min temperature (used for temp_range variable), and rainfall for regression
USDA NASS Crop Condition Progress	Contains sentiment score data by crop and year	Collected via USDA API → Stored clean data as JSON → generated weekly and annual CSVs for analysis	**5,823 rows** for weekly sentiment scores from USDA averaged for annual sentiment scores	Contains sentiment scale corresponding to commodity by year for sentiment chart

Features in Analysis

Name	Scale/Units	Description	Analysis Used In
Crop Yield	Bushels/Acre	Expresses ratio of volume of crop yield produced by total acres farmed in given state & year	Multivariate Regression
Commodities	{Corn, Wheat, Soybeans}	Analysis done on categorical, subgroups of crops	Multivariate Regression
Average Temperature	Degrees Fahrenheit	Mean temperature recorded per state and year	Multivariate Regression
Average Temperature Range/ Volatility	Degrees Fahrenheit	Calculated from subtracting mean max temp from mean min temp to get mean temp range per state and year	Multivariate Regression
Total Annual Precipitation	Inches	Total precipitation fall in inches per state and year	Multivariate Regression
Sentiment Score	Condition Scale: {Excellent:1, Good:0.5, Fair: 0, Poor: -0.5, Very Poor: -1}	Categorical scale of crop condition recorded weekly	Sentiment Analysis

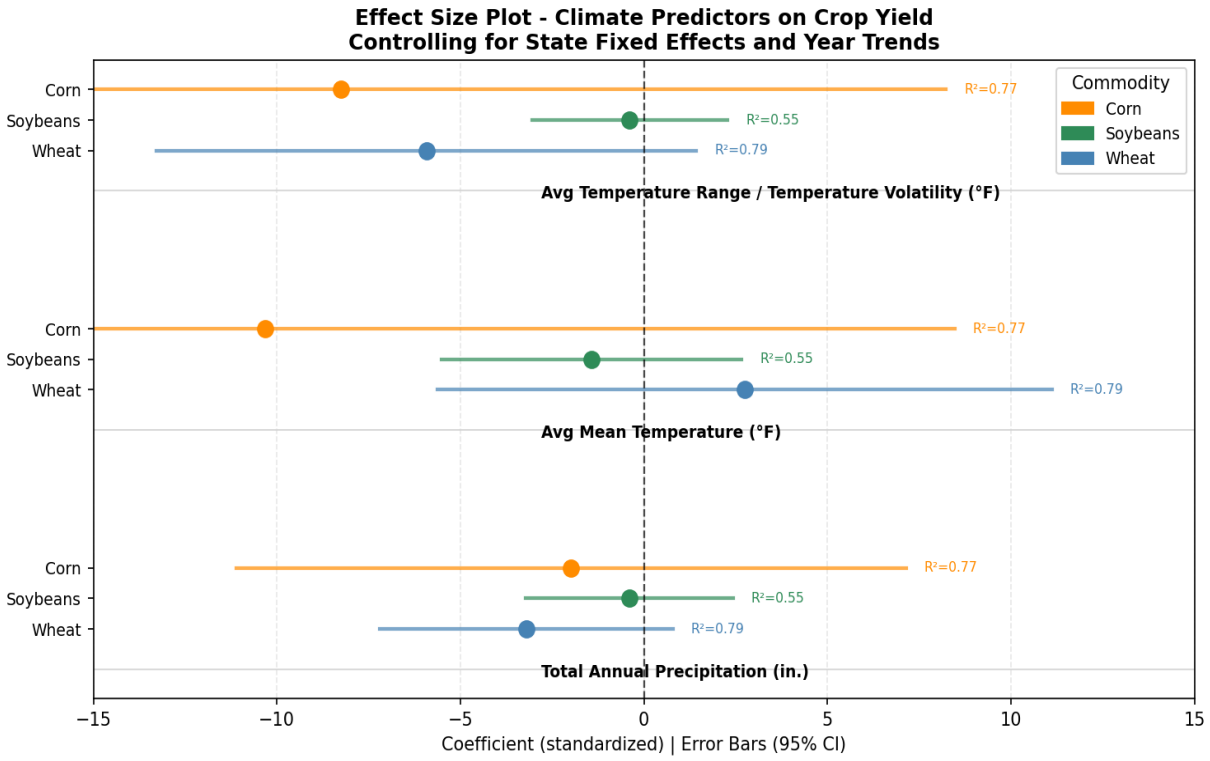
Regression Findings

Key Insights

Temperature volatility has a negative effect on crop yield for both **corn** and **wheat**, but close to zero for **soybeans** indicating climate resilience

Average mean temperature has a strong negative effect on **corn** yield. **Corn** has the highest yield loss from extreme heat stress compared to other key crops because of high sensitivity (Zhong 2022).

All crops have negative standardized coefficients for annual precipitation signaling that excessive rainfall and flooding over the past 15 years may have affected overall crop yield



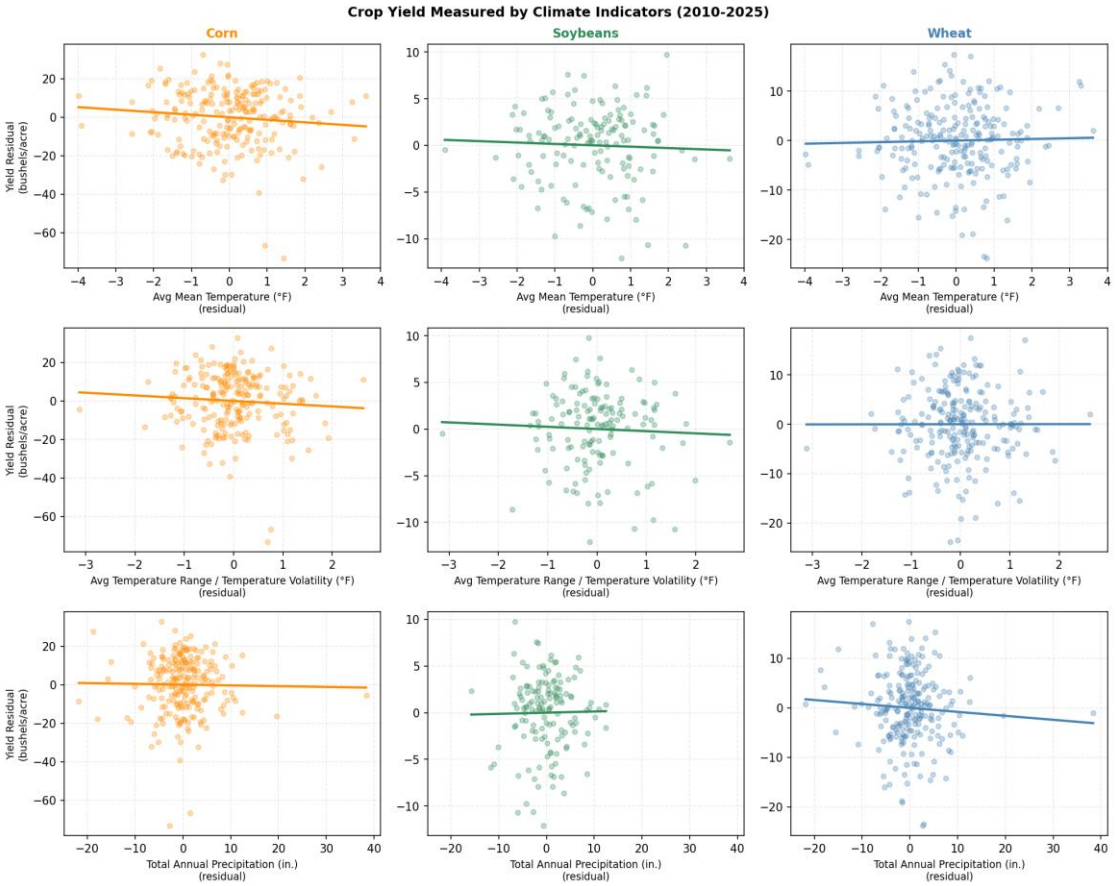
This plot holds state fixed effects and year trends constant to isolate predictors' impact on yield

Regression Findings

Key Insights

With higher than predicted average mean temperature in a year, **corn** yield falls below expected output controlling for geography and technology advancements

Higher annual precipitation is correlated with a stronger negative yield for **wheat**, aligning with literature that excessive rainfall may cause rotted or modly grain and weeds (Gandia et al., 2021)

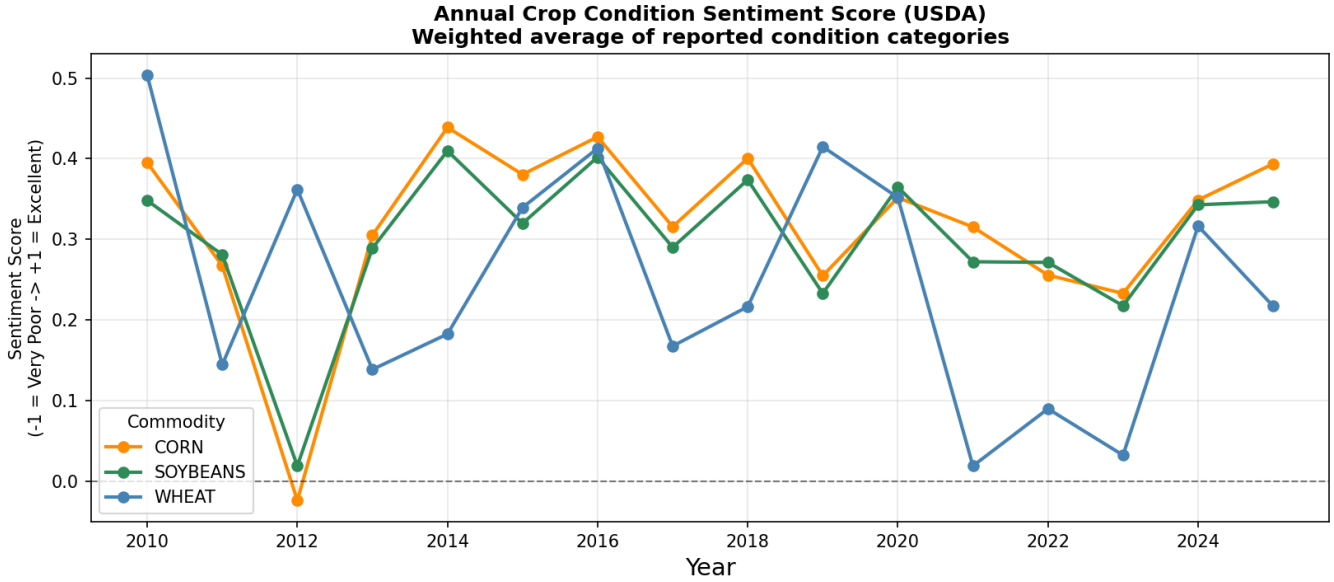


These graphs hold state fixed effects and year trends constant to isolate predictors' impact on yield

Sentiment Analysis

The sentiment chart shows that **Corn** and **Soybeans** had a near negative average sentiment score from farmers, aligning with the 2012 moderate/extreme drought in the Midwest that affected 86% of the Corn and Soybean Belt region (NOAA, 2012).

After 2020, farmer sentiment for **Wheat** had a steep decline until 2023. This corresponds with 69% of winter wheat production experiencing drought in 2022 and a 37% reduction in wheat production due to precipitation loss in 2023 (U.S. Wheat Associates, 2023).



Challenges

1. Joining USDA and NOAA data sources since they came in different data formats (text files and CSVs) and aggregating data on a yearly level for merging. NOAA data also gets ovewritten every month, so had to figure out using regex for retrieval.

2. Avoiding multicollinearity of avg. max and min temperature. So, learning how to transform these predictors into a temperature range variable. Also needed to learn how to control for state and year in model to control for major geographical differences.

3. Had difficulty on how to accurately set up regression and sentiment analysis from skitlearn and visualizing with seaborn packages due to unfamiliarity with coding structure.

Thank you!